I’m a little sleepy. Well, I have the solution for that. We’re about to talk to sleep expert Matt Walker. Matt Walker? Yeah. Second time on the show.

Mm-hmm. Looking forward to that. I’m looking forward to a nap now. Coming up on StarTalk Special Edition.

This is StarTalk Special Edition, which means I got Gary O’Reilly here. Hi, Neil. All right. Add Chuck. Nice, baby.

That’s right. And we’ve been calling you Lord of Nice, not just Lord Nice. The Lord of Nice. Lord of Nice. Very nice.

So, Gary, we’re talking about sleep today, I’m told. Oh, yes. People spend, like, a third of their life doing just that. I know. If you’re lucky. If you’re lucky.

If you’re lucky. If you don’t have kids, you spend a third of your life. I mean, I sleep like absolute dog crap. So… I don’t know what that means, but… Can you guess?

I haven’t checked how much dog crap sleeps. So, what do you got for the show today? All right, sleep. As we’ve decided, it’s an inscapable part of our daily lives. It’s weird. It can transport us into a psychotic state.

It can leave and leave us experiencing time dilation. It can heal us. Yes, it’s… Anyway, the last time we had a sleep expert on, our audience were very, very impressed. And so, it’s been a couple of years in the making, but we have brought him back, along with, Neil, our audience’s curiosity.

So, we’re going to have some questions from our audience tagged on to our conversation. Nice. Nice. Nice. Okay. Yeah.

So, let us introduce our returning champion. Sleep champion? How returning? Yes. Professor Matt Walker. Welcome back to Start Talk.

It’s lovely to be back, Gents. Thank you so much for having me. Excellent. And last we caught up with you, you see Berkeley just tearing down the house, but you’ve got new digs. What’s the latest with you?

I have. I have shifted. I am now a professor of neuroscience and biomedical engineering at UT Dallas at the center for brain health. And here I'm starting a new center for advanced brain performance and sleep innovation. So, I thought I would try and mix it up a little bit and…

Wait, wait, wait. Advanced brain performance while you're sleeping or is that a separate thing from sleeping? It's 24 hours. So, we're going to try to augment the human brain both when we are wide awake as well as when we are sleeping. Wow.

Okay. Now, you're spooking me out here. All right. Look at that. The word augment, I think, is what spooks me a little bit. Yeah.

It's going to make sleep zombies. Okay. Don't we have those already? Yeah. Exactly. Now, you've also been a TED talker and you've got the podcast, the Matt Walker podcast.

You couldn't be a little more creative than that? I know. It took years to come to that creative decision. I mean, there was the marketing, the research dollars spent just to come up with that task. You know? I mean, it was…

I thought it was borderline genius. There's so much going on in that title, isn't there, only the depth of it, which reflects my own personality. I am desperately shallow, but I have a very deep surface, at least to say. There you go. No.

You're okay. I'm desperately shallow with a deep surface. See, what he did there? I like Christine behind me. So, what we want to know is what, you know, it's been a couple of years, what's some new sleep research over those couple of years?

And what a lingering question that I've just always had is why do we need sleep at all? Because it's too early in the show to start off with that question. Well, you went there. I kind of went there. Because I think about aliens, if they visit us, and then you say, excuse me, Mr. Alien, I have to lie down horizontally and go semi-comatose for one-third of Earth's rotation.

I'll be back to you. The alien might just look at you like, what the hell is wrong with you? And so, could you reflect on this so that when the alien comes, I'll have an excuse? I'll give you the best question I've ever received in my entire career. Let me try. So, I love the way that you frame it, because imagine the birth of your first child and

the doctor comes into the room, they say, congratulations, healthy baby girl. Everything looks good. We've done all of the tests. And they're walking out the room. And just before they exit, they turn around to you, maybe like the alien, and they say, there is just one thing, though, from this moment forward, and for the rest of your child's

entire life, they are going to lapse into a state of non-consciousness. It's going to look like death, but don't worry, it's reversible. And whilst their body lies still, their mind will be filled with bizarre hallucinogenic experiences. And we have absolutely no idea why this happens. Good luck.

Take care. All the best. Well, my response would be, I'll have what the baby's having. Sign me up for that. That sounds really damn good. But it's so perplexing because from an evolutionary perspective, sleep is the most idiotic of

all behaviors. You're not finding a mate. You're not reproducing. You're not foraging for food. You're not caring for your young. And you're vulnerable to predation.

On any one of those five grounds, sleep should have been strongly selected against in the cause of evolution unless it serves an absolutely vital set of functions. And 50 years ago, to your question, we used to say, well, why do we sleep? And the best answer we had was we sleep to cure sleepiness, which is the idiotic equivalent of saying that we eat to cure hunger. Now, you eat for all sorts of physiological reasons.

Now, 50 years later, based on the wealth of the data, we've had to upend the question and ask, is there any major physiological system of your body or is there any operation of your mind that isn't wonderfully enhanced by sleep when you get it or demonstrably impaired when you don't get enough? And so far, the answer seems to be no. And we can dive into the different functions.

But I would say to your first part of the question, what's kind of fresh and new or just amazing, I would say two things. The first we've been looking at what we call genetic short sleepers. So these are people who, based on the evidence, can get away with as little as about five to six hours of sleep and show no impairment in their brain or their body, whereas everyone else, it's a nosedive like a dart into the ground in terms of functions.

I fear I may be one of those, by the way, Matt, because I'm completely content with five and a half hours of sleep. Yeah, but you can also sleep anywhere at any time. That's true. Yeah. Ah, that's a giveaway, actually.

Don't keep describing because then he's going to have a word for it. Yeah. And I worry when any time to have a word for something I do. It's called a Tyson nap. But keep going. I was going to say, we call it chronic sleep deprivation.

But I mean, if you cannot fall asleep rapidly almost anywhere, sometimes we wonder, is there a lingering sleep debt somewhere in the system where you have such a pressure for sleep that literally you can sleep, you know, within a heartbeat? That tends to want to bring a few sort of warning bells. But I mean, most people like you, Neil, would say, I wonder if I'm one of those sleepers, just to put it in context, statistically, based on the evidence, you're more likely

to be struck by lightning in your lifetime than you are to be one of these genetic short sleepers, the probability of which is very small. Ah, have you been struck by lightning? No, he didn't say both would happen to you. He's comparing two separate statistical facts. Maybe that's the progression.

You get struck by lightning, and then you can't go to sleep anymore. That's it. It's cured. That is not what the man said. OK, Matt, continue. That's what we call augmenting the human brain with a bolt of lightning.

Yeah. And there's at least one movie on that, a phenomenon with John Travolta. Yeah. Yes, he's hit by lightning, and he turns completely brilliant.

Get out. And he can predict earthquakes and move stuff with his hands and things.

But we're actually just coming back to it. We're doing a mini version of that. We've actually developed a headband, which inserts a small amount of voltage into the brain. And it's so small, you typically don't feel it, but it has a measurable impact on your brainwaves.

And what we've managed to do is create a stimulation protocol where you just wear it for 10 minutes before bed. And it's like a child on a swing. They're moving their feet, and nothing happens. But if you stop pushing them and giving them some momentum, at some point you can stop pushing them, and they keep swinging.

And that's what we're doing with the brain. We're sort of singing in time with your electrical brainwaves, and we're amplifying those brainwaves. It's almost like electrically fertilizing the soil on your prefrontal cortex, so that when you go into sleep, you germinate more powerful, deep, slow brainwaves. So it's quite far away from John Travolta's experience in terms of the voltage magnitude, but not dissimilar.

But coming back to the short sleepers, sorry, I keep losing track. What this tells us is something absolutely remarkable. Not just that this exists and there's a genetic modification for it, but it means that somewhere along the way, Mother Nature has figured out how to zip file sleep. She's been able to essentially compress eight down into six. And if we can start to find out the mechanisms, and we're digging into this now, we think

you've got some good understanding as to how these genetic sort of mutants are able to accomplish the shorter sleep. What that means is that maybe think about, you know, a world of CRISPR and gene editing. Could it be that at some point, everyone selects to be a genetic short sleeper? Let me ask you this, Matt, based on what you just said. Is it possible that the genetic anomaly, this mutation-Stop calling me a mutation. Is it possible that the Tyson mutant gene-He's turned him into an X-Man. The most obscure X-Man ever. Tyson. What's your power?

Watch. I just slept eight hours. Okay. Okay, enough. That's so cruel to you. I am so sorry.

I have no respect. Is it possible that when you find this anomaly that you may not be able to make that a reality for every other person, because sometimes genes are expressed in sets, not just one particular gene, especially when it comes to mutations like that. Yeah. So the nice thing is that we've been able to identify exactly which specific gene.

In fact, right now, there are four of these genetic short sleeping genes. If you want to select from the sort of catalog menu that I'm sure is going to be on offer anytime in the next 10 years, maybe go for what's called the DEC2 gene, D-E-C2 gene. It's one of the shorter sleeping genes, genetic mutations. But yes, you don't have to run into that problem because of the selectivity and specificity of the gene identified.

Dude. You can go right in there. Are you serious? I mean, okay. He's a professor, of course. I'm trying to resist him.

Hey, Matt, just all I'm saying, I'm going to email you offline. I need in on this. I mean, do you know how much money you're sending on? Like people don't, like you don't need stimulants, you don't need, like you're actually getting what you need in the amount of time that you're supposed to get it. You just shorten what the brain, the time the brain needs to regenerate for that coming

day. That's insane. Exactly. Amazing. As a concept. Yeah.

Absolutely. Do we take this to the nth degree and say, well, we can reprogram our brains to need no sleep at all? Ooh. Is that, is this the sort of, is that too far out or is that achievable? That's too weird.

It's interesting concept. Would you want to stay up 24 hours a day, seven days a week if you could? Now believe me, I used to do meth, so I've done it. But thank you for laughing at that, Matt, because these two are just as silent as church masses right now. Tumbleweeds.

I can see them across the screen right now. We're not laughing at your meth jokes, Yuck. I'm sorry. He's in Texas now, too. He gets to reference tumbleweeds. I know.

You see? That's how they work. But would I want to? I don't know. There's something magical about the state of A, the deep sense of relaxation and restoration that I feel when I come out of deep sleep.

And also, dreaming, I would be so sad to lose the optionality of dreaming, so I don't think I would. Hey, this is Kevin the Somalia, and I support StarTalk on Patreon. You're listening to StarTalk with Neil deGrasse Tyson. Before we go to Q&A, because this is a queries episode, right? Yes, it is.

Yes, it is. From our Patreon members. And because you know they're thinking about sleep, so there's the value of sleep as a restorative force on our bodies and minds, perhaps. But then, what is the value of dreaming? And is that any different from day dreaming?

Oh, yes, it is based on the architecture, based on the patterns of brain activity. They're not quite the same thing. But to your first part of the question, what does dreaming do? And we have to be clear, what does dreaming do above and beyond the stage of sleep that it comes from? So we have too many types of sleep, non-rem sleep and REM sleep, and most dreaming comes

from REM sleep or rapid eye movement sleep. And it's not just whether you have REM sleep. REM sleep has a series of different physiological functions, but dreaming on top of REM sleep also has selective benefits, and there are at least two. The first is that dreaming provides a form of informational alchemy, which is that every morning when you wake up, having gone through dreaming, you wake up with a revised, mind-wide

web of associations, because it's during dreaming that we take all of the information that we've recently learned and we collide it with this back catalog of information that we've stored up across a lifetime. And we don't start to make the obvious connections during dreaming. That is what we do when we're awake. Dreaming is biased towards building the more distant, non-obvious associations.

So the whole thing is they say, sleep on it. Yeah, right. That's where you get that phrase from, sleep on it. Correct. How many times has someone ever said to you, you know, Neil, you should really stay awake on a problem.

Right. No one tells you that. No one says that. They tell you to sleep on a problem. And that's exactly one of the functions. It's almost, it's like group therapy for memories.

Everyone gets a name badge and then dream sleep gathers all of these sort of people and it forces you to speak to the people at the back of the room that you don't think you've got any association with whatsoever. But when you do, you find it's a distant, non-obvious connection. Because when you start to fuse things together, that shouldn't normally go together. But when they do, they cause a marked advance in evolutionary fitness.

That sounds like the basis of biological creativity. And so that's one of the functions of dreaming. Well, it would have to be, especially when you have associations like Zebra and Women's Lounge Array smoking a cigar while drinking a martini. How do you actually relate that to anything in your life? Tell me that.

Not that I had that dream. Was it a friend? Not that I had that dream. The friend of Freddie Ors. The friend of Freddie Ors. Exactly.

I'm just saying that. You know, there's very few sleep questions that I've ever left me dumbfounded. However. So let's get to the questions. We have a bunch of questions here. With Matt's explanation about dreaming, then to take sleep out of our lives completely

wouldn't make any sense at all, would it? No, especially because I mentioned to us that the second function is perhaps even more relevant to one of keeping it, which is that dreaming provides a form of overnight therapy. Dreaming is emotional first aid and it's during dreaming at night when we take difficult, sometimes painful experiences and dreaming acts like a nocturnal soothing balm and it just takes the sharp edges off those painful, difficult memories so that you come back the next day and you feel better about that experience. I got one. I'll tell you what happened to me. The night of September 11th, 2001, I'm four blocks from the collapse of the towers. Yeah. And so I have to get my family out.

I was going to say you were still. You go ahead. No, no. It was it was traumatic. You had. I maintained composure, right?

But it was traumatic. Yeah. I had to get my two kids. One is nine months old. One is four and a half. And I collect my wife in midtown because we live downtown and we went up to my parents

in Westchester, New York, OK, very safe distance from and they were running trains in one direction. Yeah. They weren't taking people back in. OK, right. So the reason why I'm saying this is at the end of the day, I went to sleep and I woke up 11 hours later and I was astonished because it didn't feel like 11 hours, but I'm echoing

your point. My brain had to process what the hell just happened. Exactly. And that was more than a day's worth of content, I would say. And so is it just the easy explanation here that my brain wasn't ready for me to wake up yet?

Is it just processing all this disparate content that didn't happen in a typical day? We see this time and time again, whether it's whether you're learning vast amounts of information that requires memory consolidation and the cementing of new information into the architecture of the brain or that you've had difficult emotional experiences. There is a selective drive and a demand from your brain and your body to get more sleep, to see if it can reconcile the weight of the kind of burden of life that you've given

it that prior day. And so it fits exactly, I think, with your experience. You know, there's a beautiful and an American entrepreneur, E. Joseph Costman, had a wonderful phrase. He once said that the best bridge between despair and hope is a good night of sleep. And I would say that in the past 20 years of studying sleep and mental health, we have

not been able to discover a single psychiatric condition in which sleep is normal. And I think that tells me everything that I need to know about the intimate relationship between your sleep health and your mental health. We're saying your body knows. It explains a lot about me. Thank you, Matt.

You're very welcome. I'm understanding myself much better now. It's just a therapy session here. Shall we dive into it? Send us the bill for that. Let's get some picture on questions.

OK, Matt. Sit up straight. Here we go. This is a great name. Ronald Nigel McWilliams from Selma, Alabama. Thank you for what you do.

You're welcome. Thank you for being one of our Patreon group. Let's put the glass on. If the brain can generate a fully immersive reality during dreams without external input, what does that reveal about waking consciousness? Are dreams just neural noise or do they expose that what we call reality is always a controlled

construction of the brain with sleep revealing the machinery more clearly? That's quite a deep one. That's quite a big one. Just to understand the question, is it that the brain is processing your awake state? I was going to say your woke state, but that's got baggage today. Your awakened state.

And does your brain know that that is an objective reality versus what it dreams about? Is that where he's trying to get at here? I think what he's saying is that all of reality is a perception of our brain. So when we go to sleep, does that kind of perceptive reality then just become neural noise or does it serve a purpose to reveal the machinery more clearly? That was good.

That was good, Chuck. Great question. Wonderful synopsis. I would say what this question to me reveals is an uncomfortable truth about your dreams, which is that your brain can build an entire universe. So space, time, faces, emotions, memory, narrative, coherent, out of nothing.

There's no light hitting your eyes. There's no sound. There's no world. You're cut off from reality, but in the dark of this thing that we call our skull, it is manufacturing a fully convincing reality and it does it every single night. And that has an implication because if the brain can do it when essentially the doors

are closed, what makes you so sure that what you experience when your eyes are open? Don't mess with us like that now. Stop messing with us. I'm going to have to go further because the reality is it actually isn't any different than that. What we call reality right now is also a construction.

It's a model that your brain is building behind your eyes. And the only difference is that the waking perception is almost like a controlled hallucination, one that's essentially tethered to time, second by second by second. And that information is streaming from your senses. That's the only difference between your waking consciousness and your dreaming consciousness. My name is Neo.

Yeah, you're coming in and out of the matrix there just in case you didn't know. Programs, hacking programs. But no, so it's a wonderful demonstration that I think there are these hard boundaries that we think of that are actually much more porous than we imagine that there's this thing called the dreaming state and the waking state. They're very similar in some ways, except that dreaming has a feature to it that really

gives it away, which is that you have no logical, rational reasoning or control. And that's why you start to believe things which are not there. So you hallucinate, you are delusional, you get confused about time, place in person, you're suffering from disorientation, and then how wonderful you wake up in the morning and you forget most if not all of that dream experience. So you're suffering from amnesia.

If you were to experience any one of those five things when you're awake, you'd be seeking some psychological treatment. But for reasons that we now understand based on the patterns of brain activity when you go into dreaming, this seems to be a normal biological and psychological process. And the principal reason that dreams are the way that they are is because the thing that makes us most human, which is called the prefrontal cortex,

that part of the brain, unlike the rest of the brain, the brain lights up when you go into REM sleep. In fact, some parts of your brain are 30% more active than when you're awake, which is done except for the prefrontal cortex. That goes offline. So now you have these emotional centers, these memory centers, the visual centers, the motoric

centers of the brain, they're all active. But the prison guards, they've left the building and the prisoners are running amok. And that's the beauty of dream sleep. There is no sort of CEO in control with top-down regulation. There's no logical rational control.

The inmates are running the asylum. Basically, that's exactly what it is. And we believe that there's good utility to that. That's where you get kind of the loosey-goosey. Everything starts to get connected together in these wacky, strange ways. No wonder you can solve problems in your sleep in a way that you can't with this very rigid

way of thinking that is the waking state. Wow. Now you've scared me. However, that's not it. So your dreams were the world's first psychologist, first of all, right? They were.

Exactly. It is your therapist. Yes, your therapist. In a therapist. In a therapist. Well, what you just said there, Matt actually answered Aaron Gannon's question as well.

So Aaron, thank you for your question. So let's move on to Sam, who says, greetings, Dr. Tyson, Lord Nice, Gary, and Mr. Walker. My name is Sam, and I'm writing to you from the home of Casey Jones, Jackson, Tennessee. Is that right? Look at that. He says, I recently read a study by Laura Lewis from the Boston University about cerebral

spinal fluid washing the brain during sleep. That question is, Dr. Walker, could the waves of cerebral spinal fluid that wash through the brain during sleep actually be generating the dream state rather than the other way around? And if so, what does that mean for what we think dreams are for? Thank you for your time and consideration.

Live long and prosper. Well, thank you, Sam, you big nerd. All right. Well, I mean, just generally, what is this cerebral spinal fluid and its relation to sleep because Sam has already read the study. We have not.

So maybe if you start there, we'll be able to unfold an answer for him. Does it have any validity? Well, that's that, you know, whatever you know about it, does it have an immense amount of lid? I think it's one of the most stunning new discoveries in the field of sleep. Wow.

We used to think that the brain didn't have its own cleansing system. The body did, which is called the limb fatigue system, and we've known that for several centuries in fact. But then a scientist at the University of Rochester, Macon Nettigard, made an incredible discovery working in mice. First, you found that the brain does have a cleansing system.

It's called the glimphatic system named after the cells that make it up called glial cells. The second thing that she discovered is that this cleansing system, which uses cerebral spinal fluid, this sort of, it's almost like the WD-40 of the brain, it uses the cerebral spinal, terrible, uses cerebral spinal fluid to essentially flush out all of the metabolic detritus that builds up during the waking day. And that was her second discovery.

This glimphatic washing cycle is not switched on in high flow volume across the 24 hour period. It's only when you go into sleep and specifically deep sleep. And this will come back to Sam's question. It's only when you go into deep sleep that this cleansing system kicks into high gear. So it's good night's sleep clean.

It's a power cleanse for the brain. The final thing that she discovered, which makes it really relevant for disease states, is that two of the pieces of metabolic detritus that the glimphatic system cleanses when you go into deep sleep are beta amyloid and tau protein. These are the two underlying culprits of Alzheimer's disease, and it explained why for now over 10 years, and we've done some of these early studies, we were seeing a very strong relationship

between people who were getting less than six hours of sleep across the lifespan and a markedly high risk for developing Alzheimer's. But it was an association in search of a mechanistic causal answer, and she discovered it. And we've now actually developed a silent MRI, science MRI scanning protocol, and we're having people sleep inside the MRI scanner. And we can now see this pulsing cleansing system in real time, as people are sleeping

in the scan. It's absolutely beautiful. But it actually comes back to Sam's question though, which is, it turns out that dreaming comes from principally a different stage of sleep. It comes from rapid eye movement sleep, whereas this glimphatic system happens during deep non-rem sleep.

So the two are disconnected. But while we're here, because we keep referencing stages of sleep. I think it'd be nice to have a baseline, and you give us the actual stages of sleep, and how many cycles would that happen throughout the night, these stages themselves. So when everyone's head hits the pillow tonight, you're going to go on this incredible roller coaster ride in and out of different stages of sleep.

And as I mentioned, we have two main types of sleep, non-rem sleep and REM sleep. Now non-rem sleep has been further subdivided into four separate stages, unimaginatively called stages one through four, increasing in their depth. So stages three and four, that's that really deep restorative sleep. So what happens is that when you fall asleep, you go into the light stages of sleep, and then you go down into the deeper stages of non-rem sleep.

And after about 70 or 80 minutes, you'll start to rise back up, and you'll pop up and you'll have a short REM sleep period. And then you go down again, down into non-rem sleep, and then up into REM sleep. And you do that, on average, about 90 minutes is each cycle's length. What changes, however, is the ratio of non-rem to REM within that 90 minute cycle as you move across the night.

And what I mean by that is, in the first half of the night, the majority of those 90 minute cycles are comprised of lots of deep non-rem sleep and very little REM sleep. But as you push through to the second half of the night, that tita-tata balance, that shifts, and now more of that 90 minute cycle is comprised of REM sleep, dreaming sleep, and very little deep sleep. And that's the reason if you've been woken up from a phone call at sort of one o'clock in the morning, you almost never remember a dream. But if it's late in the morning, sort of five, six, seven in the morning, and you get woken up, typically that's when you'll remember a dream, because you're moving into the REM sleep-rich phase of your night of sleep. Does that make some sense? Absolutely.

So by the way, I discovered these cycles when I was in college, when I was studying my sleep cycles. Right. And so 90 minutes is a bang-on unit of time for me. If I don't have time to do five and a half hours, 90 minutes, I don't even need an alarm. So you get an hour and a half to go back out.

Correct. And I have several of these throughout the night, which are hugely useful when you were portioning the times of day where you got to get stuff done versus when you're going to go to sleep. But I would say, though, to your point, Neil, I love the fact that you can almost get that quartz-like precision.

There is a bit of a myth, though, that I say, on average, it's 90 minutes. It can be as little as 70 minutes, and it can be as long as 120 minutes, depending on who you're studying. Got it. I'm not 120. If I wake up after 120, I'm groggy, but if I wake up after 90, I'm not.

Perfect, because you've gone down into the deep state of sleep again. I discover this empirically is my point. Yeah. You know how I did it? Did I say this the last time? Are we going back to my meth joke?

No. Please don't. So what I did was, I drank, like, you know, half a gallon of water. That's the Native American alarm clock. And then I went to sleep, and then throughout the night, I woke up to go pee, but my body woke itself up to go pee, and those were bang on to the cycles.

When I'm in deep REM, it doesn't wake me up. So I made note of those timings, and then that's how I used to use it. So that's just me. I don't know. I'm the one who was hit by lightning, apparently. Well, I've done the same thing, except it's just my failing prostate.

Oh, seven days. Did you need to share? No, what I love about it is, you spoke a little bit about, if I go past that nine to 90-minute mark, and I then wake maybe an hour and… It doesn't work. 45 minutes.

It doesn't work. It doesn't work. Because, as I said, we've gone down to that other cycle, and when you come out of deep non-REM sleep, that's very unusual. Your brain never naturally does that, and what happens is that when you go into deep sleep, a large amount of cerebral space, particularly your frontal cortex and other areas, they

all go power down, and it's almost as though when you come out of deep non-REM sleep, you try to go from the basement to the penthouse, and you get stuck on the 13th floor, and you get this groggyness. It's almost like a sleep hangover effect, and it's what we call sleep inertia. When you come out of deep sleep, it takes much longer for the brain, like a classic car engine, for example.

It just needs time to gradually warm up before it gets to operating temperature, which uses, I was just about to go, I love it, you know, it's impatico. You and me, Neil. The value of this to me is, often I take very early airplane flights, so I got to leave home at four to catch a six o'clock plane. If I'm up till midnight or something, or if I'm working, I can combine two one-and-a-half

hours, that's three hours, I'm not going to go to sleep at midnight and wake up at four, because that's out of sync, so I will stay up another hour so that I will wake up on the cycle at the right time. At the right time. And so sometimes it's staying up another 15 minutes or 20 minutes or a half hour. And so this works with precision.

Thank you. What's the term used? Quartz. Quartz precision. I love that. What's like precision?

But I would say that people sort of have this, you've got to be a bit careful for these kind of gadgets that say, I'll wake you up at the perfect moment, because they all assume the average 90-minute cycle, but the problem is, no one is the average. Well, I'm the average, but it's the… Well, except the sublimely extraordinary, but yet average, can you all do that? Okay.

Do better now. Next question. All right, we've got Kenny McFarlane, he's really precious about pronunciation. Kenny's from Washington, DC, and thank you, Kenny, for sharing your own personal story here. We're going to get the thoughts of Matt.

As a kid, I experienced a number of parasomnias, sleepwalking, sleep terrors, nightmare disorders, sleep paralysis. Still some occur. Why do we grow out of these conditions? Why do some seem to go away and others continue to be consistent? Does our brain chemistry change?

And that allows for them to cease? Or better, what does experiencing parasomnias tell us about our brain chemistry? Thanks for taking the question. You're welcome. Kenny's sites himself as an all too lucid dreamer. Oh, wow.

So much going on in the question. I'll try and be efficient, but what Kenny's describing are the parasomnias. These are things that happen in and around sleep, but aren't quite sleep itself. So sleepwalking, sleeptalking, night terrors. People often think that if someone is sleepwalking or they go over to the refrigerator, they open the door, they kind of have something that they must be dreaming and they're just acting

out their dreams. That's actually not the case. When you have a sleepwalking or sleeptalking event, sleepeating event, what's happening is that your brain is in the deepest stages of non-room sleep and we can record it and we can see people walking around enacting these what seem like complex behaviors, but the brainwaves look like deep, slow brainwave sleep.

And what they're doing is simply acting out, fairly routinized, just simple kind of more sort of basic motor behaviors. And if you wake people up and typically you shouldn't, you should just sort of make sure that they're safe, but just let them go through it. But if you wake them up and you say, what was going through your mind just before I woke you up?

And they'll say, nothing. My mind was blank. And the reason is because they were in deep dreamless sleep. They weren't having a dream. So going to the refrigerator is different from asking to start a game of chess, right? Well, you said that they tend to do routine, open the window.

Exactly. Yeah. Something that is very repeatable. Yes. Yeah. These are, this is not sort of, well, some people have enacted things such as they get

in their car and they actually drive and, but most of this happens much lower down the sort of the food chain of basic behaviors. However, sleep paralysis is something a little bit different. And I will get to the Kenny's question as to why does it change as we get older? Why do we have less of these events? Sleep paralysis is a beautiful demonstration of how normally two things go lockstep hand

in hand, which is that as we're waking up, well, let me go back when you go into dream sleep, your brain paralyzes your body so your mind can dream safely. So every time you go into dreaming, there's a signal sent down your spinal cord and it paralyzes all of the voluntary muscles, not the involuntary ones that keep you breathing, but the voluntary muscles that you use to kind of reach over and grab a glass. All of those are paralyzed, but normally when we're waking up out of a dream, your consciousness

as it's starting to return to you, as that's starting to come up, the paralysis is gradually released in perfect lockstep so that you wake up and you can move and you can talk and you don't even think twice about it. Sleep paralysis happens where your consciousness gets a little bit ahead of its skis and you're starting to wake up, but you've not been released from the paralysis. So you can't move, you can't talk, but you're aware of your surroundings and typically this

happens with a sense of something else or someone else being in the room and it turns out that if you look at all of the characteristics of sleep paralysis, they often explain quite a large majority of so-called alien abductions, correct? Because when was the last time anyone here heard this new story? They were saying, today in downtown Dallas, Jimmy was just whooshed away by aliens. We were sitting around the board table and all of a sudden whooshed, Jimmy was gone.

It never happens like that. It’s usually at night when we’re in bed, people describe that the aliens injected them with a paralyzing agent. They couldn’t move, they couldn’t talk, they couldn’t fight them off. This is classic sleep paralysis and so this is one of the other features, it’s not sleep walking or sleep talking, it’s a distinctly different sort of component of this broad

collection of things that we call the parasomias. Why do these things, however, change as we get older? Well, it turns out that the boundaries between sort of waking consciousness and these sleep events is quite porous when we’re young and in part it’s because of the prefrontal cortex developing, the more that your prefrontal cortex develops, the more capable your brain is of creating hard boundaries between being awake and being asleep.

But when we’re younger, because of that prefrontal cortex not yet being fully matured, the states have been very much kind of awake and conscious versus being very much asleep and non-conscious. They’re a little bit more sort of free flowing and that’s where you can have intrusions of essentially wakefulness into sleeping states and that’s what we call parasomias. Now as we get older, we can still maintain some of these things and I think Kenny is describing that he still has some of these things too.

That’s fine, it doesn’t mean that there’s anything necessarily wrong with you as long as it’s not causing safe issues. It just means that your prefrontal cortex never developed, Kenny. No, not saying that at all. You have no executive function, Kenny, I’m sorry to tell you, Kenny. That’s what Matt has told you.

Kenny, you’re fine. Kenny, you are absolutely golden. As long as there’s no safety concerns, you are A-OK. However, wouldn’t it depend on how old Kenny is? Because from what I understand, the prefrontal cortex and executive function is the very last thing to harden in the development of the brain.

It is and that’s why essentially these things can even be extended into late teenage years. I would say to Kenny’s question too, I think he mentioned something about the chemistry of the brain. It’s not really the chemistry of the brain, it’s the structural architecture of the brain that’s changing and that’s why those episodes will start to dissipate the further into adulthood we go because of the structural development rather than the chemistry itself.

By the way, stress and sleep deprivation, both of those markedly increase the probability of those parasomnia clustering events happening just as an FYI. Interesting. All right, here we go. This is Anna Ivinko and Anna says, hi, Dr. Tyson, Dr. Walker, Lord Nice and Gary, greetings from Croatia.

My name is Ivinko and I have a question about the importance of sleep scheduling. I work as a doctor in the emergency medicine department and my shifts are as follows. 12 hours, day shift, 7A to 7P, night shift the next day, 7P to 7A and then one day of rest. Often I work two day shifts and then a night shift for two nights and then a day shift. How do I preserve my circadian rhythm and therefore my physical and mental health on this job?

I really like my job, but I also want to live long, long time fan and recent patriarch on patron and yeah. So why don’t you go ahead, Matt, and tell Ivinko that she is screwed. No, no, no. Would you stop that? She needs a union.

That’s what she needs. I’m a union. You need sleep. You need a union, girl. You need sleep. Anna, Anna, Anna.

Oh, my goodness. Okay. For start off, what is the circadian rhythm? Circadian rhythm is simply a rhythm that lasts approximately one day. So circadillas, sort of essentially the circling of one day of time. So that’s your circadian rhythm.

We all have one and it’s almost like this sinusoidal sort of wave where we are more awake throughout the day because we are a diurnal species and then our biology ramps down as we go throughout the night and then we step and repeat it. And you have a circadian rhythm. It’s driven by a 24-hour clock in your brain and it just beats out that beautiful 24-hour rhythmic pattern.

Okay. Briefly, I heard that they took people underground and took away other watches and had them just live their own understanding of day and night. And when they did this, they lived 25-hour days. They didn't live a 24-hour day, none of them. So we might be fighting our own rhythms by staying with the sun.

Are you familiar with that study? Was it replicated? What was that about? It is. It's been one of the most replicated findings and rather than think of it that we're fighting our biology, it's simply that our biology has a little bit of sort of error margin in

it and it simply needs the sun, which has been the most consistent signal since the dawn of time for a rhythmic circadian rhythm. And it latches on to the sun and the sun acts like a set of fingers on a wristwatch. And it simply just kind of tweaks your 24-hour rhythm back to precisely 24 hours each and every day. Okay.

So we're not fighting a 24-hour day. No. It just tunes us to it and without the sun, our circadian rhythm gets sloppy. It gets a little rayward. It does. It runs on average for most people.

It runs about 24 hours and about 11 minutes. So it's different for different people, but on average, that's the… So you drift forward in time by a little bit each and every day. So we need a leap rhythm. A leap rhythm. And we need a sort of leap rhythm.

From what I understand, blind people have a problem with this as well. They can develop a problem with this as well. Yes, they do. Depending on the nature of the blindness, if it's at the level of the retina, there are cells inside of your eye whose principal job is not to construct vision. They have no interest in vision.

All they simply are doing, they are light meters. And they are there to tell your brain, do you have light or do you have dark? And if you lose those cells, your brain is never able to receive that signal of light. And you can start to have difficulties in your circadian rhythm. But to come back to Anna's question with night shift work, firstly, I love the fact that Anna loves her work and that she wants a longer life.

That’s exactly the right instinct, of course. But I should probably be honest and trying to be practical at the same time. The hard truth is that rotating night shifts are tough on your circadian biology. And there are health consequences that have been associated with it. In fact, the World Health Organization some years ago, and they repeated this mandate just recently, they classified any form of nighttime shift work as a probable carcinogen

because of the relationship between a disruption of your sleep, wake rhythms, and the relationship with cancer. And that’s not meant to frighten you. It’s meant just for us to earn the respect for the risk that you take by giving us the opportunity that if I have an appendicitis at 3 a.m. in the morning, there’s someone there to save my life, which is unbelievably incredible.

So firstly, thank you for that. But because your body runs on that natural 24-hour sort of cycle, when you fight your biology, you typically lose. And the way you know you’ve lost is disease and sickness. And so what to do, what can we sort of help here? There’s probably a few things.

The first I would say is that direction matters, that if you are a shift worker and you have to do these rotating shifts, try to shift clockwise with the clock. So do sort of a morning shift, then an evening shift, then a night shift, then a morning shift. Some people don’t have the flexibility, but if you are someone who is creating these shifts or you have the optionality, try to shift forward in time.

And your biology is much harder to go in reverse. The second thing I would say to Anna is be careful of your drive home. That is the most dangerous part of your shift. And so if you can try and get a lift home, try and take public service, whatever it is, just try to be careful because a tired driver is based on the evidence, not dissimilar to a drunk driver if you look at some of the data.

The next thing I would say is try to defend your daytime with as much nighttime-ness signalling as possible. What do I mean? When you get home, make sure that your home is blackout curtains, you are ensconced in darkness, get the room cold because your body is expecting a temperature drop at night, eye mask, earplugs, and then try to take a little bit of melatonin about two to three

hours before you expect to try to sleep during the day. You can also take a hot bath or a shower about an hour before bed. Why would you do that? Your brain and your body need to drop their core temperature by about one degree Celsius, two to three degrees Fahrenheit, to fall asleep and stay asleep. It's the reason that you'll always find it easier to fall asleep in a room that's too

cold than too hot because too cold, it's the right temperature direction. So why the hot bath or shower? You think, well, I get out of the bath and I'm all kind of toasty and that's why I sleep. Well, it's the opposite, that when you get into the hot bath or shower, it draws all of the blood from that's trapped at the core of your body out to the surface. It's almost like a snake charmer that you're drawing the hot blood out.

And when you get out of the bath, all of the heat that's trapped in the body is radiated out because it's been lifted to the surface and your core body temperature actually plummets and that can help you because your body temperature. You're tricking your body into thinking that it's nighttime. It's nighttime. Exactly.

That's very cool. So I would say those are the things that you could try to do. Again, just be careful of the drive home and then make your home when you get home like a cave, cold, dark, quiet. So, Matt, real quick, I just want to say, because I think this is important from my own personal experience and maybe you could speak to it.

I had a job where I had to get up at three o'clock in the morning and I was still doing comedy at night. So I wasn't getting any sleep. OK. And I've been very open about the mental illness issues that I have. And all I can say is what you just talked about, anyone who suffers from mental

illness of any kind, whether it's depression, bipolarism or anything along the along those lines, the sleep is so very important. And I went into a very dark period because I just wasn't getting sleep. And so the sleep health that you just described is really important for those of you who suffer from mental mental health issues. I want to salute that flag every single time you hoist it

because sleep is it’s not part of the equation. It is the foundation on which your mental health sits. And wow, there’s this almost you can’t clean your way out of insufficient sleep. You can’t exercise your way out of insufficient sleep and trying to protect your sleep when you are you when you know you are vulnerable from a mental health context, or if you know someone who is suffering from mental health

conditions, try to see if you can help stabilize someone’s sleep. That means the circadian rhythm, giving them bright, lovely, some exposure to daylight in the first 60 minutes when they wake up. That actually sets like an hourglass turned over a timer to instigate the release of a healthy dose of melatonin at night that can stabilize your sleep rhythms. And then when it’s nighttime, digital detox,

try to turn down almost all of the lights in the home, try to disengage the mind. But I cannot just emphasize more than what you just said. It is such an essential ingredient. And when it goes away, goodness, can we become vulnerable? And we become vulnerable very quickly based on the studies that we’ve done. Digital detox means put away the phone and the tablet.

That’s right. I’d rather die.

I’m sorry, I was channeling my children. Time for a couple more. John Geffen from Winwood PA. I have been fascinated by the recent discussions of consciousness in StarTalk. So what does our understanding of non-REM sleep tell us about consciousness? Can consciousness really turn off during sleep or might it continue

throughout and just be blocked from our memory when we wake? Oh, what a good question. So in other words, if you’re asleep and you’re not an REM, what is your state of consciousness? From everything that we can tell, you are absent consciousness, meaning that if I wake you up from deep non-REM sleep and I say, what was going through your mind? Nothing. Nothing will come to mind. Wake you up out of light non-REM sleep. Often you can have these sort of little mini dreams of LITE dreaming, as it were. But when you go into REM sleep, that's full blown dreaming. But there was a there was a beautiful kind of twist in that story, which is, well, is it just simply that my memory is blocked?

Now, that actually is the case for most of us with REM sleep dreaming. Because as I mentioned earlier, you have all of these Florida experiences and you typically forget a lot of your dreams. But I don't know if that's true or not. Yes, and we think we understand why it is that dreams dissolve so quickly when you wake up, it comes back to the prefrontal cortex,

because it's your prefrontal cortex that is capturing all of the information and then sending it to another part of your brain called the hippocampus, which sits on the either left and right side of your brain. And the hippocampus packages that those experiences as memories. But it needs the prefrontal cortex as a spotlight to direct those memories into the packets.

And if you don't have a prefrontal cortex that's active, which you don't when you're dreaming, it's very difficult, we believe, to package and send those memories into storage. I've got a wacky theory that we don't forget our dreams. In fact, I would argue we may remember every single one of our dreams. And it's the difference between what I call accessibility versus availability.

How many times have you had that experience where you woke up and you think, I was dreaming, I know I was dreaming. And as hard, the harder you try, the further you push it away and you just give up. And then two days later, you get into your car and you see maybe a pack of chewing gum and it's a cue

that all of a sudden unlocks that full dream memory from several days ago. What that tells me is that your brain is storing the memory. But what you've lost at the moment of waking up is the IP address for that memory. So the memory has been stored and is available. It's just that you've lost the connection for its accessibility.

So availability, accessibility, which means that if that's true, that we can have these spontaneous, cued recollections of dreams days later. What if we store all of our dreams and we know for a fact that there's a lot of information that sits below the radar of consciousness that is changing our human behavior and our decisions and our actions. What that leads to is a bit of a mind bending possibility

that all of our dreams are stored throughout our life. And those dreams are constantly shaping and changing the decisions and the beliefs and the thoughts that we have in our waking life. And we're completely unaware of it. Wow. Yes. That, first of all, that's just nuts.

I mean, I don't mean like nuts as in like you're a quack. I mean, it's nuts as in to think about how many people have said that. But yeah, but no, I'm saying it's nuts to think about the implications of what you just said, you know, but it definitely works for creativity. I can speak to that personally because I have had dreams that I wake up and I'm like, I can't remember.

And I know that they're like, I wrote a joke or I something happened. But then I'll be sitting down doing something completely unrelated at a different time. And that dream will pop in my head and I'm like, oh, man, there it is. And I'm like, oh, that's the joke. And then I'll tell the joke and people will be like, you suck.

No, I'm joking. I'm joking. No, yeah, but it's useful. It's useful is what I'm saying. Yeah. And isn't that I mean, and there's a couple of experiments that I think I could design to try to prove that that's the case.

And then we could get into the whole world of, OK, if you can understand where memories are stored during dreaming, could you start to manipulate those memories during the dream state itself? Can you alter memories or can you selectively decide which memories that your sleeping brain brings up and remembers and which it wants to selectively forget?

We're starting to get to the stage now where we've developed experimental tools where almost like a playlist, I can select what it is from your waking day that you decide to remember and what you lose. The even deeper implication of what you just said is if that's the case, then maybe all memories are stored that way. And if that is so and you're able to manipulate it, Alzheimer's patients,

you would be able to find a way to raise their memories like the way we dust for fingerprints or something. You know, you'd be able to raise those memories and they wouldn't suffer. They might still have Alzheimer's, but they'll still be able to access the memories. And the question is, is Alzheimer's a condition of availability but impaired accessibility?

And the sad part is that because of the neural degeneration that happens, particularly in the memory centers, we think that, in fact, those memories are no longer available at all. Oh, so it's like a damaged. It's like a corrupted file. Yeah, it's like a corrupted file.

Follows there, but you can't get to it. That's a shame. Good idea. Yeah, tragic. I tried. I tried to help you people. Can we take one more question, but we got to do it like in 30 seconds. Rapid fire. Yep. Come on, Walker.

OK, Brian Edair, I was wondering how the astronauts on Artemis 2 sleep on the shuttle. Do they have any special procedures and would we expect them to have different types of dreams or nightmares based on being in zero gravity in space? Thank you so much.

OK, then we're on the shuttle just to just to clarify that. Sophie Edair is aged 11. She's given a pass. OK, OK. I got you. By the way, great question, Sophie. It's a great question. I think it's one of the best questions of the night.

How do astronauts sleep up there? I would say, honestly, with some difficulty and a lot of Velcro, because when you're on the space station, astronauts will actually climb into a sleeping bag that is strapped on the wall. And the reason is because so they don't drift around

and start bumping into things. And there is no up or down. And as a result, you actually, because of that zero gravity, you don't need a pillow pressed against your head. So astronauts will actually describe sleeping almost like this sort of gentle floating experience.

And some of them love it. Some of them never get used to it. But the real troublemaker is actually not the gravity or lack thereof. It's the sun because circling the earth, the crew will actually see a sunrise and a sunset roughly every 90 minutes. So they will see 16 sunrises and sunsets every single day.

Now, as we spoke about with your circadian rhythm, your brain carries this internal clock and it's carried it for three and a half million years. And it's being set to only one sunrise and one sunset every 24 hours. So it scrambles the brain is what you're saying. Your clock gets shouted at 16 times a day

in a completely unexpected manner. So astronauts will typically black out the windows. There were eye masks and they will keep a very strict schedule that is sort of determined to make sure that they get timed light in accordance with when their biology is wanting it. And they will get timed darkness together with a little dose of melatonin

to try to help convince their brains that it's nighttime when mission control says it is something different. And of course, they only orbited Earth a few times before they did a trans lunar injection and headed to the moon. So then they're no longer orbiting Earth because now Earth is not blocking their view of the sun,

putting them into nighttime. And so they still have to think about a rhythm because now they're just floating in space towards the moon. Yeah, any sunrise or sunset? No sunrise or sunset. No, nothing. Yeah.

Well, that's right. I'd say, though, that's probably better than 16. Like, I'd rather have no sun. Like, can you imagine somebody saying to you like, let's take a romantic stroll on the beach and watch the sunset again? Up that sunset. You know how many sunsets I've seen?

I'll give a damn about no sunset. So News Flash, Christina Koch, one of the Artemis to astronauts, came back and said she had some of the best sleep she ever had in her life. Oh, I'm just saying. Well, that's kind of sad

because you're not going back to space to sleep. Like, you're never going to find that here on Earth. Well, here's a funny one. So Tempur-Pedic, when they first came on the scene, they marketed themselves as being the material under the seat of astronauts, OK, that go into space.

And you say, wow, I want to wear sleep on what the astronauts sleep on. Except when they're in zero G, there is no pressure on the phone. There's no memory. There's no memory for the memory phone to remember. Exactly. Yeah, so it was a little, it was a little spurious there, as I saw.

Well, I hate to tell you, I have a Tempur-Pedic mattress at home. You do? Yeah, so in space, there's no bed sores. There's no half the reason why you turn at night is to change the pressure on your body.

On your body. That's correct. You're in zero G. Yeah, there is no pressure. It's just floating. I think there was a mention of, you know, what about dreams and how would that change?

Am I, we don't have actually good data on this, but my scientific hunch is that your dreams are probably not altered by anything to do with gravity. It's the emotion of being, you know, a tiny crew in a tin can, thousands and thousands and thousands of miles

from everyone that you love, looking back at the whole earth through a window. I'd wager that probably shows up somewhere in their dreams. And I would love to see the emotional kind of arc of acceptance of the monumental nature of what I'm experiencing in this moment, what this means to me.

And then the integration of all of that. Could you imagine what an absolute mind twisting? I like to imagine that they all wake up and go, I had a dream I was falling. Really? I had that same dream. They all had the dream that they were falling.

Like, yeah, guess why? All right, Matt, we got to call it quits there. It's been delight to have you back on. Absolute pleasure. Thank you. Clearly, our fan base, our Patreon fan base agrees with me that you you've taken us to places we don't typically go.

And Special Edition is all about the human condition. Yeah. Yeah. And so thanks for helping us gain some insight there. It was lovely to chat, Gents. As always, I suspect I will have some very interesting dreams tonight as a consequence of this.

And we're going to find you on your podcast, eponymously named. Is that do I say that? That's correct. Matt Walker podcast. I presume you talk about sleep topics.

I do indeed. And is there a website or a Twitter handle we can find you on? You can just go to sleepdiplomat.com. And if you want to find me on X, I am also there. Sleep Diplomat and on Instagram. I am Dr. Matt Walker, D-R-M-A-T-T-W-A-L-K-E-R.

All right, we'll find you there. All right. All right, dude. And any books you're writing? My previous book, Why We Sleep has been out for now some time. And I will simply say that there is a second one that apparently should be being written at this time.

Deadline. Now it's going to be called Why We Procrastinate.

The tagline will be Torpa hibernation and what happens when you dream about going to Mars. Anyway, that, yeah, there is a second book soon to be arriving on channel. No great piece of writing is ever finished. It only comes due. Woo.

Look at that. Give me that t-shirt, please. All right, Chuck. Always a pleasure. Always. Yeah, Brie.

Hey, Neil. In the house, Neil deGrasse Tyson and another edition of StarTalk Special Edition. This one on sleep. Thanks for joining us. Until next time, keep looking out.